

PAC learning

Lee-Ad Gottlieb

PAC Learning

- Probably Approximately Correct
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Approximately



Probably



PAC learning example

- We observe a sample of n students (drawn uniformly at random) who were accepted and rejected to a certain program
- We also know each student's psychometric score. Assume this is the sole criterion.
- Given a new random student and his psychometric score, can we guess if he will be admitted to the program?



PAC learning example

With equal probability, $1/n$

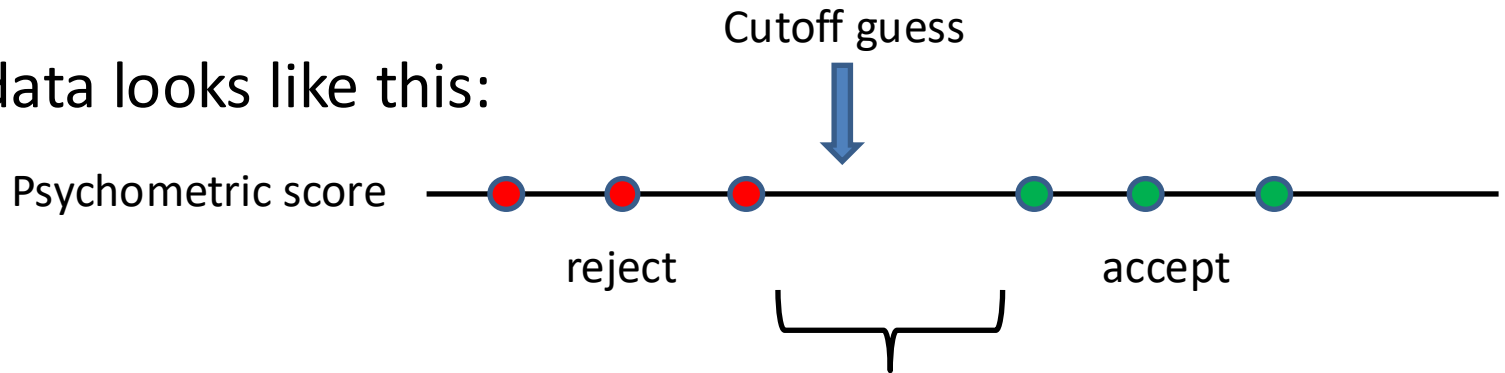
- We observe a sample of n students (drawn **uniformly** at random) who were accepted and rejected to a certain program
- We also know each student's psychometric score. Assume this is the sole criterion.
- Given a new random student and his psychometric score, can we guess if he will be **admitted** to the program?



Label: accepted / rejected, yes / no, 1/0

PAC learning example

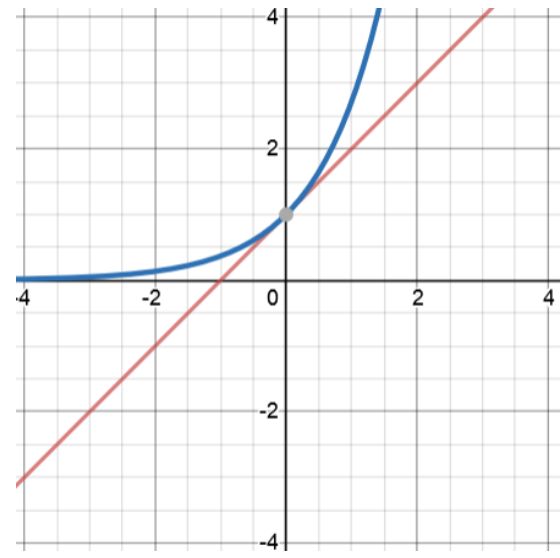
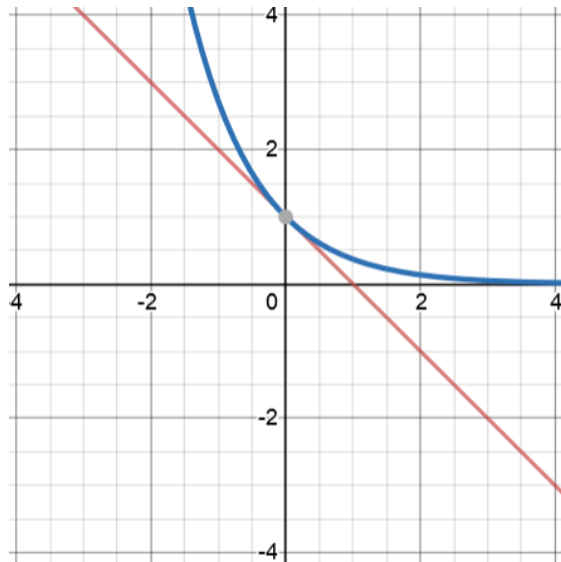
- The data looks like this:



- The true cutoff must somewhere in the middle. What learning rule should we take?
 - Guess a cutoff. Left is reject, right is accept
- Where should we put the threshold?
 - Leftmost? No false negatives.
 - Rightmost? No false positives.
 - something else?

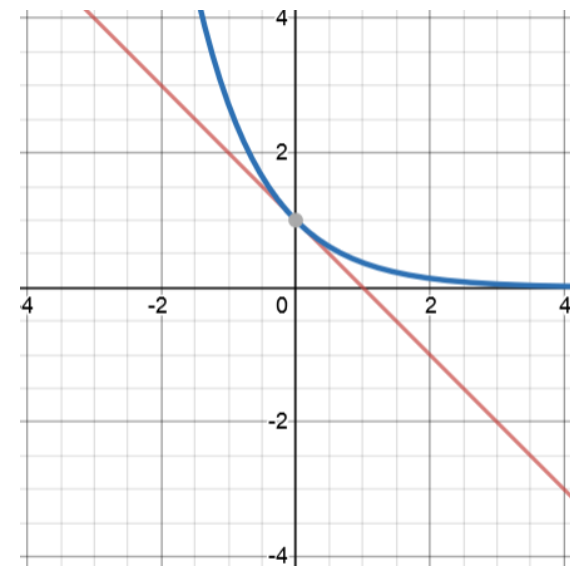
Preliminary

- Before we begin the analysis...
- Prove: $1 - x \leq e^{-x}$ for all real x
- Equivalently: $1 + x \leq e^x$ for all real x



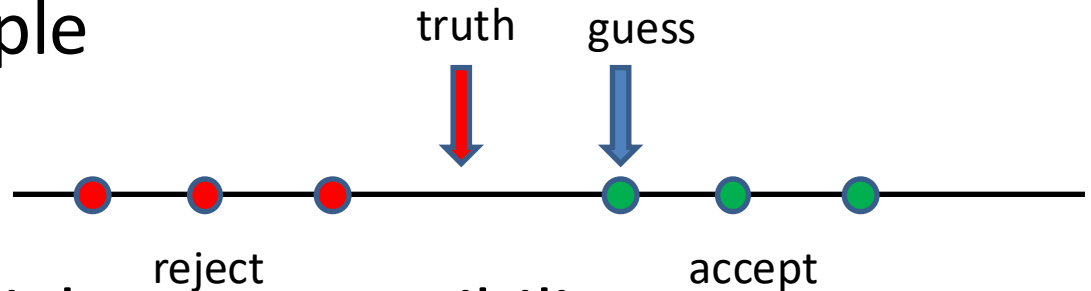
Preliminary

- Prove: $1 - x \leq e^{-x}$ for all real x
- Set $f(x) = 1 - x$ $g(x) = e^{-x}$
 $f'(x) = -1$ $g'(x) = -e^{-x}$
- Proof:
 - When $x=0$: $f(x)=g(x)=1$
 - When $x<0$: $f'(x)=-1 > g'(x)$
 - When $x>0$: $f'(x)=-1 < g'(x)$



PAC learning example

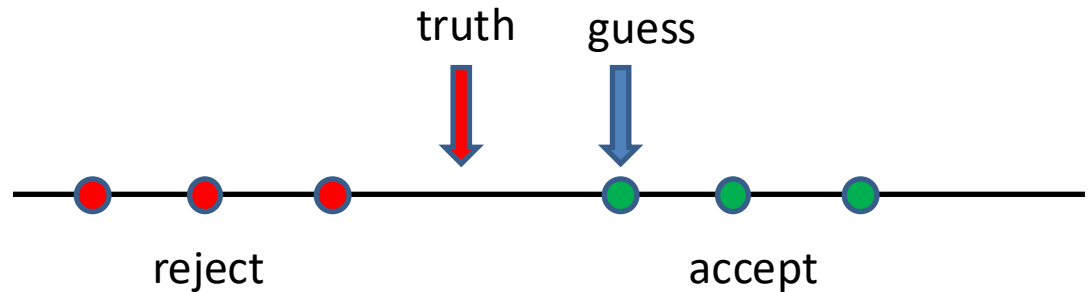
- Back to the example



- Let's choose the rightmost possibility.
 - No false positives
- What's the probability of false negative?
 - Fix ϵ, δ
 - Want a PAC statement: With probability at least $1 - \delta$, we make a mistake on at most ϵ fraction of the students

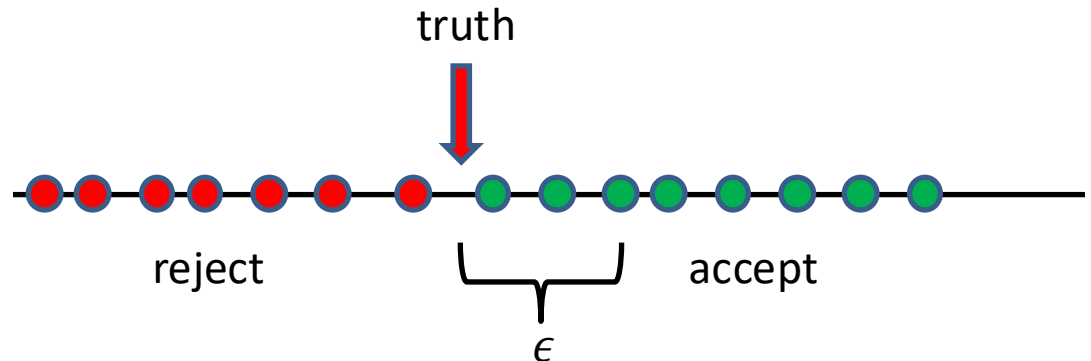
PAC learning example

- Sample:



- Same as asking: What's the probability that the sample misses an ϵ -weight near the truth?

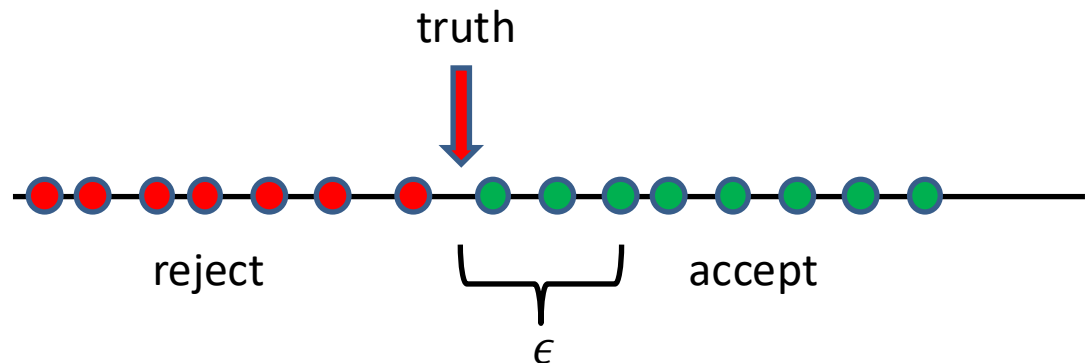
- World:



PAC learning example

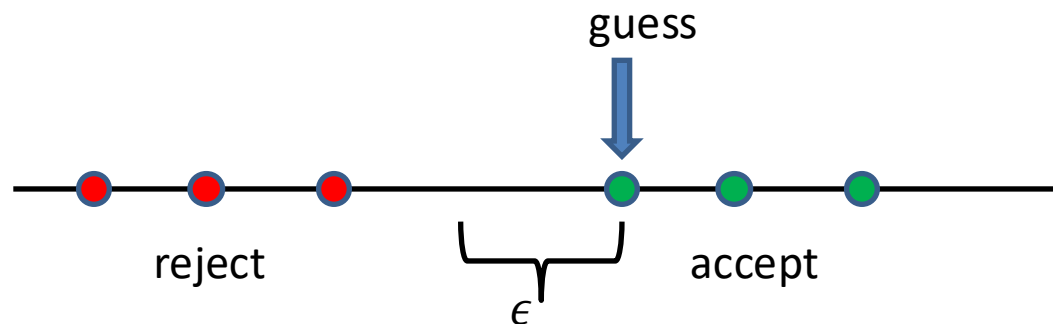
- What's the probability that the sample misses an ϵ -weight near the **truth**?
 - The probability of one sample point missing this area is pretty high, $1 - \epsilon$
 - But the probability of all **n** sample points missing this area is only $(1 - \epsilon)^n \leq e^{-\epsilon n}$
- We want the error to be less than δ , so take

$$e^{-\epsilon n} \leq \delta \quad \rightarrow \quad n \geq \frac{\ln(\frac{1}{\delta})}{\epsilon}$$



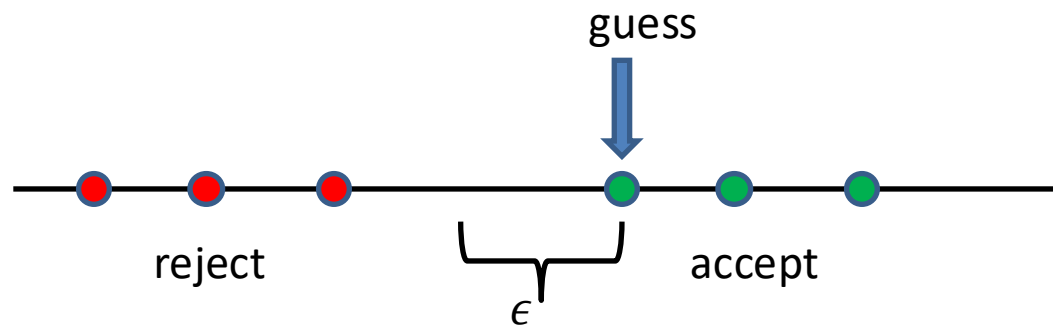
PAC learning example

- Question: Is the following different analysis also correct?
- What's the probability that the sample missed an ϵ -weight near the **guess**?
 - The probability of one sample point missing this area is pretty high, $1 - \epsilon$
 - But the probability of all **n** sample points missing this area is only $(1 - \epsilon)^n \leq e^{-\epsilon n}$
 - so take $e^{-\epsilon n} \leq \delta \quad \rightarrow \quad n \geq \frac{\ln(\frac{1}{\delta})}{\epsilon}$



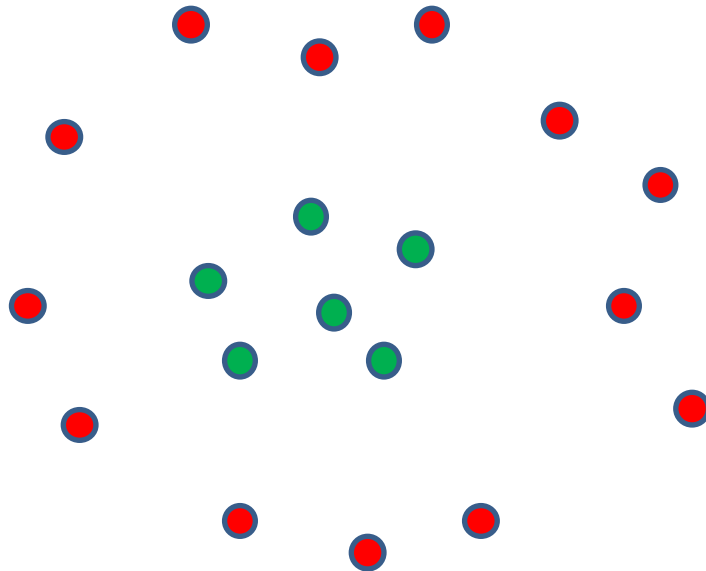
PAC learning example

- Question: Is the following different analysis also correct?
- No, guess is determined **after** the sample.
 - Sample dependent.
 - Fix: could multiply probability by number of possible guesses.
 - But this could be large.



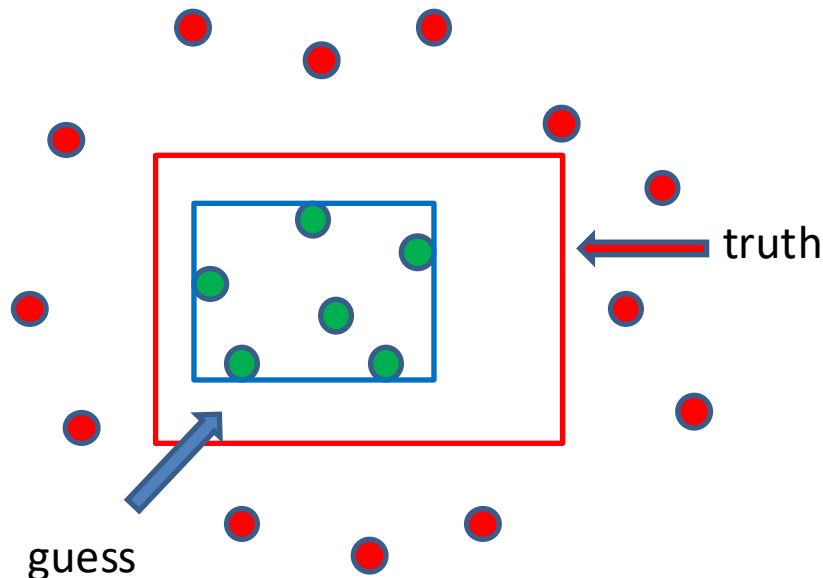
PAC learning example

- Extend the previous 1-D example to 2-D
- Let our rules be rectangles.
 - Ex: Sample of baby height, weight



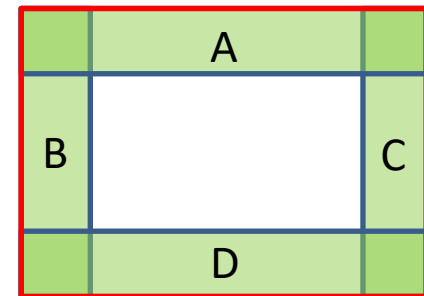
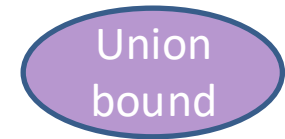
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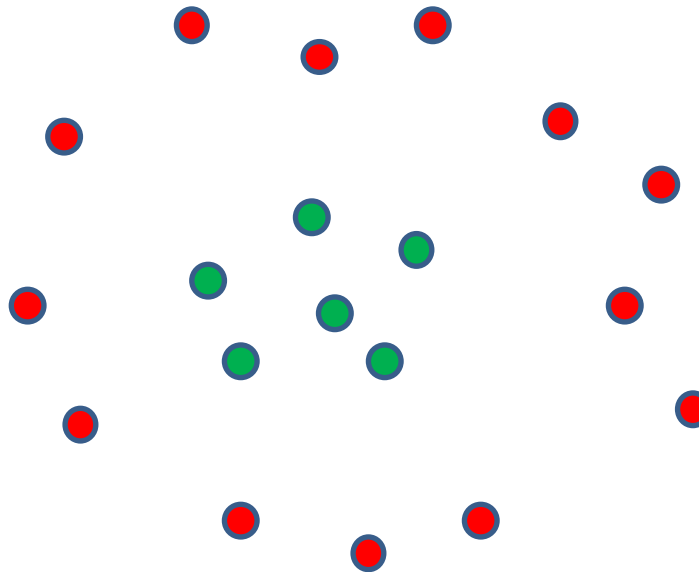
PAC learning example

- Extend the previous 1-D analysis to 2-D
- Four areas, each of weight $\frac{\epsilon}{4}$
 - Total weight $\leq \epsilon$
- Analysis:
 - $\Pr[\text{sample missed area A}] = \left(1 - \frac{\epsilon}{4}\right)^n \leq e^{-\frac{\epsilon n}{4}}$
 - $\Pr[\text{sample missed even one area of A,B,C,D}] \leq$
 $\Pr[\text{missed A}] + \Pr[\text{missed B}] + \Pr[\text{missed C}] + \Pr[\text{missed D}] \leq 4e^{-\frac{\epsilon n}{4}}$
- Take $4e^{-\frac{\epsilon n}{4}} \leq \delta \rightarrow n \geq \frac{4 \ln(\frac{4}{\delta})}{\epsilon}$
- With probability at least $1 - \delta$, no areas missed $\rightarrow$ we make a mistake on at most ϵ fraction of the babies



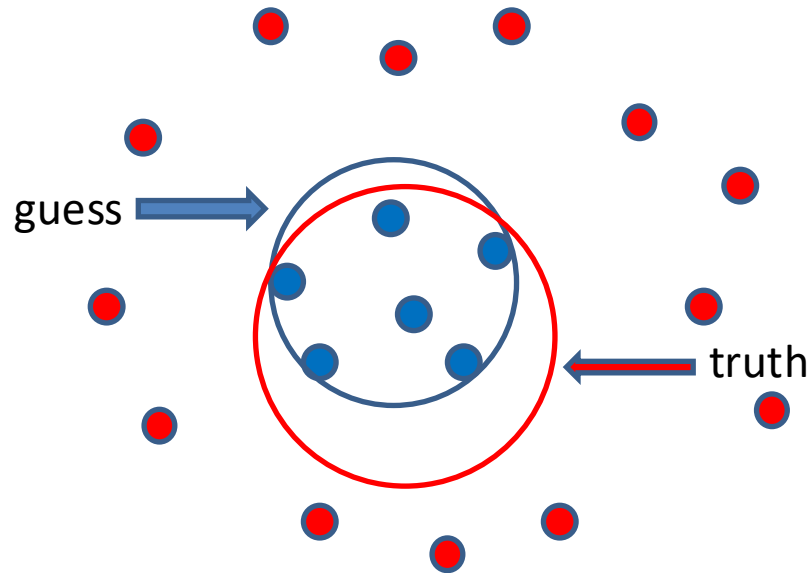
PAC learning example

- What if our rules are circles?
 - Here's our sample



PAC learning example

- What if our rules are circles?
 - Here's our sample and guess



Generalization bound

- The previous analysis fails for circles!
- If we have a **finite** set of objects (rules), we could prove the following:
 - Let X be a point set, D a distribution on X , and H be a finite set of rules mapping $X \rightarrow \{0,1\}$.
 - Let S be a random sample of **n** points $\sim D$.
 - Suppose some rule $h \in H$ is **consistent** on S (**no errors**). Then with probability at least $1 - \delta$, the **true error** (**$e(h)$**) of h on all of X is

$$e(h) \leq \frac{1}{n} (\ln |H| + \ln \frac{1}{\delta})$$

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Probability of
mistake on a new
point $x \sim D$:
 $\Pr[h(x) \neq \text{label}(x)]$

Generalization bound

- **Proof:**

- For each rule $h_i \in H$, if its **true error on X** is $e(h_i) > \epsilon$, then

$$\Pr[h_i \text{ is consistent on all of } S] \leq (1 - \epsilon)^n$$

- $\Pr[\exists h \in H : e(h) > \epsilon \cap h \text{ consistent with } S]$

$$= \Pr[[h_1 \text{ consistent with } S \mid e(h_1) > \epsilon] \cup$$

$$[h_2 \text{ consistent with } S \mid e(h_2) > \epsilon] \cup \dots]$$

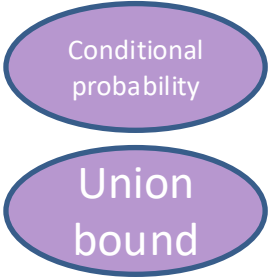
$$\leq \sum_i \Pr[h_i \text{ consistent with } S \mid e(h_i) > \epsilon]$$

$$\leq |H| (1 - \epsilon)^n$$

$$\leq |H| e^{-\epsilon n}$$

- Take $|H| e^{-\epsilon n} \leq \delta \rightarrow \epsilon \geq \frac{1}{n} (\ln |H| + \ln \frac{1}{\delta})$

- So the probability of finding a hypothesis with true error at least $\frac{1}{n} (\ln |H| + \ln \frac{1}{\delta})$ but consistent with the sample is at most δ
- If we found a consistent hypothesis, then with probability $1 - \delta$ it must have true error less than this



Conditional
probability

Union
bound

Generalization bound

- What if the rule h is not consistent with the sample?
 - Say it has **empirical error** $\bar{e}(h)$ on the sample.
- We can still prove the following:
 - Let X be a point set, D a distribution on X , and H be a finite set of rules mapping $X \rightarrow \{0,1\}$.
 - Let S be a random sample of n points $\sim D$.
 - Suppose some rule $h \in H$ has **empirical error** $\bar{e}(h)$. Then with probability at least $1 - \delta$, the **true error** of h on all of X is

$$e(h) \leq \bar{e}(h) + \sqrt{\frac{\ln 2|H| + \ln \frac{1}{\delta}}{2n}}$$

Generalization bound

- What if the rule h is not consistent with the sample?
 - Say it has **empirical error $\bar{e}(h)$** on the sample.
- We can still prove the following:
 - Let X be a point set, D a distribution on X and H be a finite set of rules mapping $X \rightarrow \{0, 1\}$.
 - Let S be a random sample of size n drawn from D .
 - Suppose some rule $h \in H$ has empirical error $\bar{e}(h)$. Then with probability at least $1 - \delta$, the **true error** of h on all of X is

Percentage of sample points on which a mistake was made

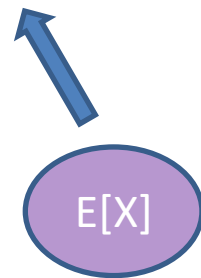
$$e(h) \leq \bar{e}(h) + \sqrt{\frac{\ln 2|H| + \ln \frac{1}{\delta}}{2n}}$$

Hoeffding's inequality

- Let **B** be a Bernoulli $\{0,1\}$ distribution with parameter p .
- Draw n random independently distributed trials $t_i \sim B$, and let $X = \frac{1}{n} \sum t_i$ be their empirical mean
- Then $\Pr[|X-p| > \epsilon] < 2e^{-2n\epsilon^2}$

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Generalization bound

- Fix a rule $h \in H$
- For each sample point $x \in S$, let the error be $err(x) = 1_{h(x) \neq label(x)}$.
- Then $\bar{e}(h) = \frac{1}{n} \sum_{x \in S} err(x)$
- So now the points with their error labels are Bernoulli $\{0,1\}$ variables.
- Hoeffding: For a single rule h ,
$$\Pr[|\bar{e}(h) - e(h)| \geq \epsilon] \leq 2e^{-2n\epsilon^2}$$

Generalization bound

- Hoeffding: For a single rule h ,

$$\Pr[|\bar{e}(h) - e(h)| \geq \epsilon] \leq 2e^{-2n\epsilon^2}$$

- Corollary: For any set of rules H ,

$$\Pr[\exists h \in H : |\bar{e}(h) - e(h)| \geq \epsilon]$$

$$= \Pr[|\bar{e}(h_1) - e(h_1)| \geq \epsilon \cup |\bar{e}(h_2) - e(h_2)| \geq \epsilon \cup \dots]$$

$$\leq \sum_i \Pr[|\bar{e}(h_i) - e(h_i)| \geq \epsilon]$$

$$\leq |H|2e^{-2n\epsilon^2}$$

- Set $|H|2e^{-2n\epsilon^2} \leq \delta \rightarrow \epsilon \geq \sqrt{\frac{\ln 2|H| + \ln \frac{1}{\delta}}{2n}}$

PAC learning

- But what if we have an **infinite** set of rules, like circles?
- Need more advanced tools.
 - Shattering
 - VC-dimension

PAC learning

- What can't we learn? Challenge can be:
- **Statistical**: Take the distribution which assigns each point $\{0,1\}$ with probability $\frac{1}{2}$.
 - This can't be learned.
- **Computational**: Take a distribution over graphs. We want to learn what percentage of the graphs are 3-colorable.
 - Determining whether a graph is 3-colorable is NP-hard.

